

Akshata Madavi

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Summary

MS Software Engineering student at SJSU (GPA 4.0) with 6 years of production data engineering and hands-on deep learning shipped in grad school. Trained and deployed a real-time anomaly detection model (XGBoost, 98% accuracy) on time-series sensor data with end-to-end Kafka streaming inference in ~10 seconds on GCP. Fine-tuned LLMs via LoRA (+10 pp accuracy) and built production ML pipelines processing 10M+ records. Brings the rare combination of large-scale data infrastructure depth and modern AI/ML engineering.

Education

San Jose State University, MS in Software Engineering May 2027
GPA: 4.0/4.0 | Machine Learning, Deep Learning, Recommender Systems, Software Systems Engineering
College of Engineering Pune (COEP), BS in Computer Engineering May 2018

Skills

- Languages:** Python, SQL
- Machine Learning:** PyTorch, TensorFlow, scikit-learn, XGBoost, PySpark ML, CNNs; evaluation (accuracy, precision, recall, F1, ROC-AUC)
- LLMs & GenAI:** Fine-tuning/LoRA (Qwen, LLaMA); RAG (LangChain, LlamaIndex); Vector stores (FAISS, Pinecone); Evaluation (Ragas, Promptfoo); Serving (Vertex AI, HuggingFace)
- Data & Streaming:** Kafka/KSQL, Spark, BigQuery, Airflow, Pandas, NumPy, MySQL
- MLOps & Cloud:** MLflow, Docker, GitHub Actions CI/CD, GCP (Vertex AI, BigQuery, GCS), AWS

Projects

VitalWatch – Real-Time Sensor Anomaly Detection (AI-Catalyst Hackathon 2025)

PyTorch, XGBoost, Kafka/KSQL, GCP Vertex AI, BigQuery, Twilio, Python

- Trained XGBoost on multi-channel time-series sensor data achieving **98% anomaly detection accuracy**; deployed on GCP Vertex AI.
- Built Kafka/KSQL streaming pipeline enriching flagged events via LLM with BigQuery data; delivered end-to-end alerts in **~10 seconds**.

Personal Finance Auditor – Agentic Text-to-SQL Assistant

PydanticAI, Gemini, SQLite, sqlglot, Streamlit, Logfire, Python

- Built a multi-agent pipeline: parallel LLM agents categorize bank statement merchants at ingest; a conversational agent answers plain-English spending questions via a Text-to-SQL tool loop against SQLite.
- Enforced two-layer SQL safety guardrail (sqlglot parse-level SELECT validation + read-only SQLite connection); instrumented all agent runs with Logfire traces (LLM calls, tool calls, latency, tokens); deployed via Streamlit UI.

Enterprise AI Governance Platform – LLM Fine-tuning & Monitoring

PyTorch, HuggingFace, LoRA, LangChain, FAISS, MLflow, AWS, Docker

- Fine-tuned Qwen 0.6B via LoRA; tracked experiments with MLflow improving accuracy from **87% to 97% (+10 pp)**.
- Designed LangChain + FAISS retrieval pipelines for failure pattern analysis; containerized with Docker for reproducible training and deployment.

Work Experience

Senior Data Engineer, LTIMindtree – Pune, India May 2023 – Mar 2025

- Engineered near real-time GCP/BigQuery data pipelines ingesting **~10M records** from enterprise sources, building ML-ready feature tables for downstream AI/ML models at production scale.
- Redesigned a multi-source ETL service (~30 GB) using parallel Spark jobs, cutting ingestion from **1 week to under 10 minutes**; served as the primary feature store for ML workloads.

Data Engineer, AI Adventures – Pune, India Aug 2021 – Apr 2023

- Led migration of course data from Moodle to MySQL; A/B testing and retention analysis drove a **20% revenue increase**.

Data Engineer, STL (Sterlite Technologies) – Pune, India Jun 2018 – May 2021

- Built near real-time GCP pipelines migrating Oracle, SAP, and Salesforce to BigQuery (**~1 TB**); applied DOE-based statistical analysis for manufacturing scrap detection and process optimization.
- Developed automated anomaly detection checks over sensor and telemetry outputs, reducing manual review time and catching production anomalies earlier.